CLASS 10

CHAPTER 1 : REAL NUMBERS

MATHS

Chapter Overview

- ★ Euclid Division Lemma
- ★ HCF & LCM
- ★ Fundamental Theorem of Arithmetic
- ★ Decimal Expansion of Rational Numbers
- ★ Irrational Numbers

1. REAL NUMBERS (वास्तविक संख्याएँ)

Rational Numbers और Irrational Numbers मिलकर Real Numbers कहलाते हैं।

Symbol : R

Real Numbers include

- → Natural Numbers (N)
- → Whole Numbers (W)
- → Integers (Z)
- → Rational Numbers (Q)
- → Irrational Numbers (I)

Difference Between Rational & Irrational Numbers

Rational Number	Irrational Number
$\frac{p}{q}$ form में लिखा जा सकता है।	$\frac{p}{q}$ form में नहीं लिखा जा सकता है।
Decimal terminating या recurring होता है।	Decimal non-terminating non-recurring होता है।
Examples : $\frac{2}{3}$, −7, 0.25	Examples : $\sqrt{2}$, π, $\sqrt{3}$

2. TYPES OF REAL NUMBERS

(A) Rational Numbers (परिमेय संख्या)

ऐसी संख्या जिसे $\frac{p}{q}$ के रूप में लिखा जा सके, जहाँ p, q integers हों और $q \neq 0$

Examples : $\frac{2}{3}$, −5 , 0 , $\frac{7}{8}$, 0.75

Decimal Form :
- Terminating
- Non-terminating recurring

Example : $\frac{1}{2} = 0.5$, $\frac{1}{3} = 0.333\ldots$

(B) Irrational Numbers (अपरिमेय संख्या)

ऐसी संख्या जिसे P/q के रूप में नहीं लिखा जा सके।

Examples : $\sqrt{2}$, $\sqrt{3}$, π , $\sqrt{5}$, 2.2310010001...

Decimal Form :
- Non-terminating
- Non-recurring

Example : $\sqrt{2} = 1.414213\ldots$

3. EUCLID DIVISION LEMMA

Statement : यदि a और b दो positive integers है, तो

$\boxed{a = bq + r, \quad \text{जहाँ } 0 \leq r < b}$

जहाँ q = quotient और r = remainder

Examples : Find HCF of 72 and 30 using Euclid Algorithm.

$72 = 30 \times 2 + 12 \longrightarrow (r = 12)$

$30 = 12 \times 2 + 6 \longrightarrow (r = 6)$

$12 = 6 \times 2 + 0 \longrightarrow (r = 0)$

$\therefore$ HCF = 6 (Last divisor)

Exam Tip
जब remainder = 0 हो जाए, उस समय का divisor ही HCF होता है।

4. FUNDAMENTAL THEOREM OF ARITHMETIC

Statement : हर composite number को prime numbers के product के रूप में uniquely लिखा जा सकता है।

Example : $24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$

Example : Find HCF and LCM of 36 and 60.

Prime Factorisation :

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

HCF (Smallest powers) $= 2^2 \times 3 = \boxed{12}$

LCM (Highest powers) $= 2^2 \times 3^2 \times 5 = \boxed{180}$

Important Relation

$\boxed{\text{HCF} \times \text{LCM} = \text{Product of Numbers}}$

Example : If HCF = 4 and numbers are 12 and 20

$\text{LCM} = \frac{12 \times 20}{4} = \frac{240}{4} = \boxed{60}$

5. DECIMAL EXPANSION OF RATIONAL NUMBERS

Concept : यदि किसी rational number $\frac{p}{q}$ (in lowest form) के denominator में केवल 2 और 5 या दोनों होते हैं, तो decimal expansion terminating होता है। अन्यथा non-terminating recurring होता है।

Rule (Formula Concept) :

यदि denominator = $2^m \times 5^n$
तो decimal terminating होगा।
Otherwise decimal recurring होगा।

Examples :

- Terminating Decimal
  $\frac{3}{8}$, $8 = 2^3$
  $\therefore$ Terminating
- Non-terminating Recurring Decimal
  $\frac{7}{15}$, $15 = 3 \times 5$
  $\therefore$ Recurring

Quick Trick :
- ★ Denominator में सिर्फ 2 और 5 ? → Terminating
- ★ कोई और prime factor ? → Recurring

6. IRRATIONAL NUMBERS PROOF (Example)

Prove that $\sqrt{5}$ is irrational.

Assume $\sqrt{5}$ is rational.

$\therefore \sqrt{5} = \frac{p}{q}$, where p, q are coprime integers, $q \neq 0$

Squaring both sides : $5 = \frac{p^2}{q^2} \rightarrow 5q^2 = p^2$

→ p^2 is divisible by 5 → p is divisible by 5.

Let p = 5k. Put in $5q^2 = (5k)^2 \rightarrow q^2 = 5k^2$

→ q^2 divisible by 5 → q divisible by 5.

But p and q both cannot be divisible by 5, as they are coprime. Contradiction arises.

$\therefore \sqrt{5}$ is irrational. Hence proved.

7. NUMBER LINE CONCEPT

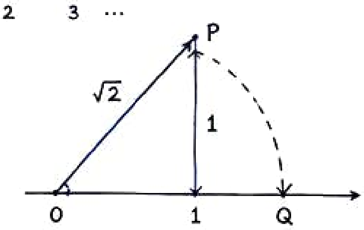
Real numbers number line पर represent किए जा सकते हैं।

... −3 −2 −1 0 1 2 3 ...

Representation of $\sqrt{2}$ on Number Line

1. 0 से 1 unit ले लो.
2. 1 पर लंब खींचकर 1 unit ऊपर point P लो.
3. OP को join करो.
4. OP की length = $\sqrt{1^2 + 1^2} = \sqrt{2}$
5. Compass से O केंद्र और OP radius लेकर number line को Q पर cut करेगा.

Q represent करता है $\sqrt{2}$.



REAL LIFE EXAMPLES

- HCF → समान size के tiles लगाना
- LCM → Traffic light timing
- Irrational Numbers → Circle measurements (π)
- Decimal Expansion → Money calculations

IMPORTANT FORMULAS

1. Euclid Division Lemma
   $\boxed{a = bq + r}$
2. HCF × LCM = Product of Numbers
3. Rational Number Form $\frac{p}{q}, q \neq 0$

SHORT TRICKS

① HCF निकालते समय Last divisor = Answer

② Denominator में सिर्फ 2 और 5 → Terminating

③ LCM → Highest power
HCF → Smallest power

COMMON MISTAKES

- ✗ q = 0 लेना
- ✗ Prime factorisation गलत करना
- ✗ Terminating और Recurring में confusion
- ✗ Proof में contradiction नहीं लिखना

LAST MINUTE REVISION

- ✓ Rational = $\frac{p}{q}$ form
- ✓ Irrational = not $\frac{p}{q}$
- ✓ Euclid Formula : a = bq + r
- ✓ Denominator only 2 & 5 → terminating
- ✓ HCF × LCM = Product
- ✓ $\sqrt{2}$, $\sqrt{3}$, π are irrational

FINAL EXAM STRATEGY : Theorem याद करो → Formulas याद करो → रोज 5 HCF questions practice करो → Decimal expansion वाले questions जरूर करो → Proof questions stepwise लिखो → Neat Presentation = Good Marks 🙂